



Sistemas Distribuidos

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TAREA 01

El mismo servicio, por REST y por gRPC

Sierra García Mariana

Grupo: 02

Facultad de Ingeniería

Evidencias de la tarea 01

```
Administrador: Windows PowerShell
PS C:\Users\sd-2027-1\entregas\sierra_mariana\tareas\t01> ls

Directorio: C:\Users\sd-2027-1\entregas\sierra_mariana\tareas\t01

Mode                LastWriteTime         Length Name
----                -
-a----          19/09/2026   10:23 p. m.         694 cliente.py
-a----          19/09/2026   10:20 p. m.         240 docker-compose.yml
-a----          19/09/2026   10:19 p. m.         183 Dockerfile
-a----          19/09/2026   10:22 p. m.         228 logica.py
-a----          19/09/2026   10:22 p. m.         221 servicio.proto
-a----          19/09/2026   10:23 p. m.        1133 servidor.py

PS C:\Users\sd-2027-1\entregas\sierra_mariana\tareas\t01> docker-compose up --build
time="2026-09-19T22:24:26-07:00" level=warning msg="C:\\Users\\sd-2027-1\\entregas\\sierra_mariana\\tare
t01\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid
potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 1.05kB 0.1s done
#1 DONE 0.1s

#2 [cliente internal] load build definition from Dockerfile
#2 transferring dockerfile: 222B 0.1s done
#2 DONE 0.1s

#3 [servidor internal] load metadata for docker.io/library/python:3.9-slim
#3 DONE 2.1s

#4 [servidor internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.1s

#5 [servidor internal] load build context
#5 DONE 0.0s

#6 [cliente 1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aa
b1acda1e13cf1731b1b
#6 resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c
f1731b1b 0.1s done
#6 resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c
f1731b1b 0.1s done
#6 ...

#5 [servidor internal] load build context
#5 transferring context: 2.96kB 0.1s done
#5 DONE 0.2s

#6 [cliente 1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aa
```

```

Administrador: Windows PowerShell

#5 DONE 0.2s

#6 [cliente 1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aa
b1acda1e13cf1731b1b
#6 sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.1s done
#6 sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.1s done
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0B / 13.88MB 0.3s
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0B / 1.29MB 0.2s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 0B / 29.78MB 0.2s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 1.05MB / 13.88MB 0.5s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 2.10MB / 13.88MB 0.6s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 4.19MB / 13.88MB 0.9s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 6.29MB / 13.88MB 1.2s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 7.34MB / 13.88MB 1.4s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 8.39MB / 13.88MB 1.5s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 9.44MB / 13.88MB 1.8s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 10.49MB / 13.88MB 2.1s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 11.53MB / 13.88MB 2.3s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 12.58MB / 13.88MB 2.6s
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB 2.8s done
#6 sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB 2.8s done
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.05MB / 1.29MB 3.3s
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 3.4s done
#6 sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 3.4s done
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 2.10MB / 29.78MB 3.8s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 4.19MB / 29.78MB 4.8s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 6.29MB / 29.78MB 5.4s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 8.39MB / 29.78MB 6.0s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 10.49MB / 29.78MB 6.3s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 12.58MB / 29.78MB 6.8s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 14.68MB / 29.78MB 7.2s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 16.78MB / 29.78MB 7.5s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 18.87MB / 29.78MB 8.0s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 20.97MB / 29.78MB 8.3s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 23.07MB / 29.78MB 8.6s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 25.17MB / 29.78MB 8.7s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 27.26MB / 29.78MB 9.0s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.36MB / 29.78MB 9.3s
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 9.4s done
#6 sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 9.4s done
#6 extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d
#6 extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 1.2s done
#6 extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d
#6 extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 1.2s done
#6 extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s done
#6 extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s done
#6 extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08
#6 extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0.7s done
#6 extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0.7s done
#6 extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.0s done
#6 DONE 12.0s

```

```
Administrador: Windows PowerShell

#7 [servidor 2/5] WORKDIR /app
#7 DONE 0.2s

#8 [servidor 3/5] COPY . .
#8 DONE 0.1s

#9 [servidor 4/5] RUN pip install flask grpcio grpcio-tools requests
#9 2.875 Collecting flask
#9 3.000   Downloading flask-3.1.3-py3-none-any.whl (103 kB)
#9 3.044   _____ 103.4/103.4 kB 2.5 MB/s eta 0:00:00
#9 5.437 Collecting grpcio
#9 5.461   Downloading grpcio-1.80.0-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (6.8 MB)
#9 6.519   _____ 6.8/6.8 MB 6.4 MB/s eta 0:00:00
#9 10.06 Collecting grpcio-tools
#9 10.08   Downloading grpcio_tools-1.80.0-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (2.7 MB)
#9 10.74   _____ 2.7/2.7 MB 4.0 MB/s eta 0:00:00
#9 10.93 Collecting requests
#9 10.96   Downloading requests-2.32.5-py3-none-any.whl (64 kB)
#9 10.99   _____ 64.7/64.7 kB 2.3 MB/s eta 0:00:00
#9 11.14 Collecting itsdangerous>=2.2.0
#9 11.19   Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#9 11.30 Collecting jinja2>=3.1.2
#9 11.33   Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
#9 11.37   _____ 134.9/134.9 kB 4.0 MB/s eta 0:00:00
#9 11.45 Collecting blinker>=1.9.0
#9 11.48   Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#9 11.65 Collecting werkzeug>=3.1.0
#9 11.68   Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
#9 11.76   _____ 226.5/226.5 kB 3.0 MB/s eta 0:00:00
#9 11.91 Collecting click>=8.1.3
#9 11.94   Downloading click-8.1.8-py3-none-any.whl (98 kB)
#9 11.97   _____ 98.2/98.2 kB 3.9 MB/s eta 0:00:00
#9 12.15 Collecting importlib-metadata>=3.6.0
#9 12.20   Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
#9 12.49 Collecting markupsafe>=2.1.1
#9 12.53   Downloading markupsafe-3.0.3-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
#9 12.62 Collecting typing-extensions~4.12
#9 12.64   Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
#9 12.67   _____ 45.6/45.6 kB 2.4 MB/s eta 0:00:00
#9 13.65 Collecting protobuf<7.0.0,>=6.31.1
#9 13.68   Downloading protobuf-6.33.6-cp39-abi3-manylinux2014_x86_64.whl (323 kB)
#9 13.76   _____ 323.4/323.4 kB 4.7 MB/s eta 0:00:00
#9 13.77 Requirement already satisfied: setuptools>=77.0.1 in /usr/local/lib/python3.9/site-packages (from g
rpcio-tools) (79.0.1)
#9 13.91 Collecting idna<4,>=2.5
#9 13.94   Downloading idna-3.20-py3-none-any.whl (69 kB)
#9 13.97   _____ 69.6/69.6 kB 2.7 MB/s eta 0:00:00
#9 14.15 Collecting urllib3<3,>=1.21.1
#9 14.18   Downloading urllib3-2.6.3-py3-none-any.whl (131 kB)
```

```
Administrador: Windows PowerShell
#9 15.03      _____ 262.7/262.7 kB 7.3 MB/s eta 0:00:00
#9 15.15 Collecting certifi>=2017.4.17
#9 15.18   Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
#9 15.20      _____ 137.0/137.0 kB 6.0 MB/s eta 0:00:00
#9 15.36 Collecting zipp>=3.20
#9 15.40   Downloading zipp-3.23.1-py3-none-any.whl (10 kB)
#9 15.67 Installing collected packages: zipp, urllib3, typing-extensions, protobuf, markupsafe, itsdangerous
, idna, click, charset_normalizer, certifi, blinker, werkzeug, requests, jinja2, importlib-metadata, grpcio,
grpcio-tools, flask
#9 17.95 Successfully installed blinker-1.9.0 certifi-2026.7.22 charset_normalizer-3.5.1 click-8.1.8 flask-3
.1.3 grpcio-1.80.0 grpcio-tools-1.80.0 idna-3.20 importlib-metadata-8.7.1 itsdangerous-2.2.0 jinja2-3.1.6 ma
rkupsafe-3.0.3 protobuf-6.33.6 requests-2.32.5 typing-extensions-4.16.0 urllib3-2.6.3 werkzeug-3.1.8 zipp-3.
23.1
#9 17.95 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour
with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io
/warnings/venv
#9 18.31
#9 18.31 [notice] A new release of pip is available: 23.0.1 -> 26.0.1
#9 18.31 [notice] To update, run: pip install --upgrade pip
#9 DONE 19.4s

#10 [cliente 5/5] RUN python -m grpc_tools.protoc -I. --python_out=. --grpc_python_out=. servicio.proto
#10 DONE 1.4s

#11 [servidor] exporting to image
#11 exporting layers
#11 exporting layers 5.2s done
#11 exporting manifest sha256:294788af3a8468cbf3b30bcffd07220d9546b62a7fcc7cab5b76d241a2c9bab3 0.0s done
#11 exporting config sha256:f0872e736453df4b62522846c54c825b0a2b88513d5a9f9b9bf46b1b7c5b0fee 0.1s done
#11 exporting attestation manifest sha256:2e0b905ab941cc10044323b04ef302847bbc47a8d40ea2f72656611a1430af29
#11 exporting attestation manifest sha256:2e0b905ab941cc10044323b04ef302847bbc47a8d40ea2f72656611a1430af29 0
.1s done
#11 exporting manifest list sha256:ed3d5a7f381bc81c55a28f862818700982c4946a01402c7edadc5d0787b088dc 0.0s don
e
#11 naming to docker.io/library/t01-servidor:latest 0.0s done
#11 unpacking to docker.io/library/t01-servidor:latest
#11 unpacking to docker.io/library/t01-servidor:latest 1.3s done
#11 DONE 6.8s

#12 [cliente] exporting to image
#12 exporting layers 5.2s done
#12 exporting manifest sha256:225ee902318dae842197dbe43ae769e01b32faf831f42669897fd31696c65fae 0.0s done
#12 exporting config sha256:ae86ffafe120eb3202f4417ba8b929b9d0167ecfaef8317b8a87014fd8c12c9a 0.1s done
#12 exporting attestation manifest sha256:da24f17c21af3df831a9088f9fb1b995d3ead7c51a334d223bfaf7f3594b8b1c 0
.1s done
#12 exporting manifest list sha256:c31f9e05b44e2dd1b9d53c078635494783659590867b024d7daaf108924b708a 0.0s don
e
#12 naming to docker.io/library/t01-cliente:latest 0.0s done
#12 unpacking to docker.io/library/t01-cliente:latest 1.2s done
#12 DONE 6.8s
```

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Administrador: Windows PowerShell
#12 exporting layers 5.2s done
#12 exporting manifest sha256:225ee902318dae842197dbe43ae769e01b32faf831f42669897fd31696c65fae 0.0s done
#12 exporting config sha256:ae86ffafe120eb3202f4417ba8b929b9d0167ecfaef8317b8a87014fd8c12c9a 0.1s done
#12 exporting attestation manifest sha256:da24f17c21af3df831a9088f9fb1b995d3ead7c51a334d223bfaf7f3594b8b1c 0.1s done
#12 exporting manifest list sha256:c31f9e05b44e2dd1b9d53c078635494783659590867b024d7daaf108924b708a 0.0s done
#12 naming to docker.io/library/t01-cliente:latest 0.0s done
#12 unpacking to docker.io/library/t01-cliente:latest 1.2s done
#12 DONE 6.8s

#13 [servidor] resolving provenance for metadata file
#13 ...

#14 [cliente] resolving provenance for metadata file
#14 DONE 0.1s

#13 [servidor] resolving provenance for metadata file
#13 DONE 0.1s
[+] up 5/5
  ✓ Image t01-servidor      Built          56.6s
  ✓ Image t01-cliente      Built          56.6s
  ✓ Network t01_default    Created        0.1s
  ✓ Container t01-servidor-1 Created        0.6s
  ✓ Container t01-cliente-1 Created         0.3s
Attaching to cliente-1, servidor-1
servidor-1 | * Serving Flask app 'servidor'
servidor-1 | * Debug mode: off
servidor-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
servidor-1 | * Running on all addresses (0.0.0.0)
servidor-1 | * Running on http://127.0.0.1:8080
servidor-1 | * Running on http://172.18.0.2:8080
servidor-1 | Press CTRL+C to quit
servidor-1 | 172.18.0.3 - - [20/Sep/2026 05:25:32] "GET /info/1 HTTP/1.1" 200 -
cliente-1 | --- PRUEBA REST ---
cliente-1 | Respuesta: {'datos': 'Informacion confidencial del usuario 1'} | Tamaño: 51 bytes
cliente-1 | --- PRUEBA gRPC ---
cliente-1 | Respuesta: Informacion confidencial del usuario 1 | Tamaño: 42 bytes
cliente-1 |
cliente-1 exited with code 0

View in Docker Desktop View Config Enable Watch Detach

```